

Unit-2

Waterfall Model Vs Spiral Model

What is a Life Cycle Model?

- A software life cycle model (or process model):
 - a descriptive and diagrammatic model of software life cycle:
 - identifies all the activities required for product development,
 - establishes a precedence ordering among the different activities,
 - Divides life cycle into phases.

Comparison between Waterfall and Spiral Model

Aspect	Waterfall Model	Spiral Model
Complexity	The Waterfall model is simple and easy.	The spiral model is a lot more complex.
Development Method	The waterfall model works in a sequential method.	While the spiral model works in the evolutionary method.
Risk Management	In the waterfall model errors or risks are identified and rectified after the completion of stages.	In the spiral model errors or risks are identified and rectified earlier.
Adoption	The waterfall model is adopted by customers.	While the spiral model is adopted by developers.

Project Size Suitability	The waterfall model is applicable for small projects.	While the Spiral model is used for large projects.
Planning	In waterfall model requirements and early stage planning is necessary.	While in spiral model requirements and early stage planning is necessary if required.
Flexibility to Change	Flexibility to change in waterfall model is Difficult.	Flexibility to change in spiral model is not Difficult.
Risk Level	There is high amount risk in waterfall model.	There is low amount risk in spiral model.
Cost	Waterfall model is comparatively inexpensive.	While cost of spiral model is very expensive.

Customer Involvement	Customer involvement is minimum in Waterfall Model	In the Spiral Model Customer involvement is high.
Maintenance	It requires least maintenance.	It requires typical maintenance.
Framework Type	It is based on linear framework type.	It is based on linear and iterative framework type.
Testing	Testing is done after the coding phase in the development life cycle.	Testing is done after the engineering phase in the development cycle.
Reusability	Reusability is extremely unlikely.	To a certain extent, reusability is possible.
Customer Control	Customer control over the administrator is very limited.	Customers have control over the administrator as compared to waterfall